SKI DATA VISUALIZATION PROJECT

DATA-3400 DATA VISUALIZATION NARRATIVE

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DATE: 2/27/25

1. QUESTION / PURPOSE

How do ski resorts in Utah compare in visitor trends, snowfall, and pricing? What type of terrain is most frequent for each resort (green, blue, or black runs) and how does that affect their popularity? How significantly does average snowfall affect prices?

2. AUDIENCE

Utah's ski resort executives, local ski/snowboarding businesses, the Utah tourism board. I would also focus on Skiers who are interested in skiing in Utah. By analyzing my visualizations, ski resort executives, local businesses, and the tourism board can make data-driven decisions to increase popularity and . Skiers should be able to look at and compare ski resorts in Utah to select the resorts that are more aligned with their price range, difficulty, and size.

3. SINGLE SENTENCE

Analyze ski resort trends across Utah, focusing on visitor growth, snowfall impact, and revenue trends, while comparing the top ski resorts.

4. DATA

I have been looking at publicly available data and reports from the National Ski Areas Association (NSAA) website, along with the information provided about each ski resort in Utah through the Ski Utah website.

5. CHART TYPES + UNITS

I'm interested in doing a line chart to show visitor trends over the past 10-20 seasons. I could potentially add in average snowfall in Utah per season as well. I also will add a stacked bar chart to show the difference in green/blue/black runs for the resorts. This will show which resorts cater to beginners or experts. Lastly, I'm considering a scatterplot to show the relationship between average snowfall and a lift ticket for some resorts. I want to see how big an effect average snowfall has on pricing.

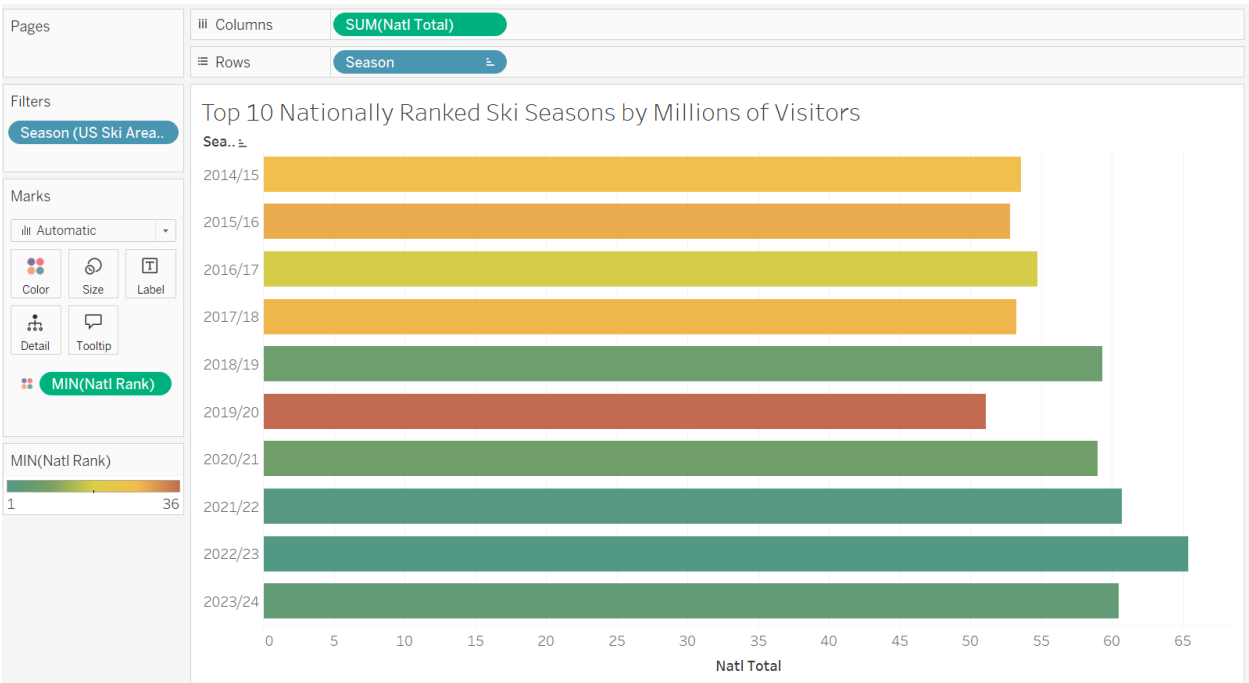
6. LOCATION IS EVERYTHING

For my spatial map, I am thinking of creating a map of the U.S. that will show either the number of ski resorts by state or the average annual visitors per region. Another potential spatial map would just show Utah and it would have the ski resorts highlighted.

7. PROCESS

02/24/25 I spent time connecting some of my data to Tableau. The data I have so far looks great! There was very little data cleansing to be done, so it was a relatively smooth process, and it was easy to merge a few sheets of my Excel spreadsheet.

02/27/25 I started playing around with some of my data and created some preliminary visualizations. So far, everything is looking good. Here is one of the charts I was testing my data with:

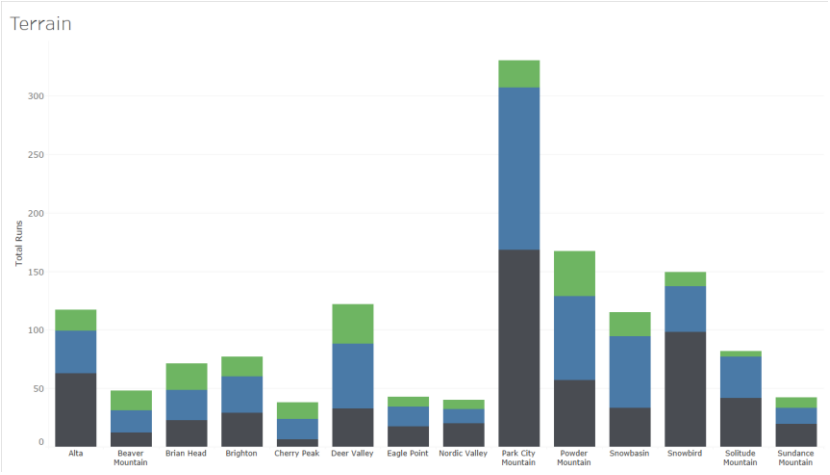


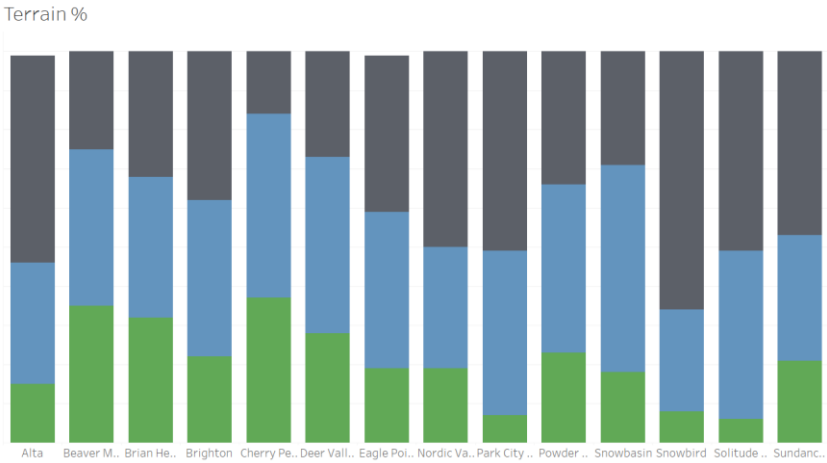
This will be helpful information for my spatial map, as it has information from the entire U.S., but I still need to clean and connect my Utah specific data for the rest of my project.

03/05/25 I was able to collect Utah specific ski resort data from the SkiUtah website and scraped pricing data for each resort (season pass price, day pass price) from the On the Snow website: https://www.onthesnow.com/utah/lift-tickets

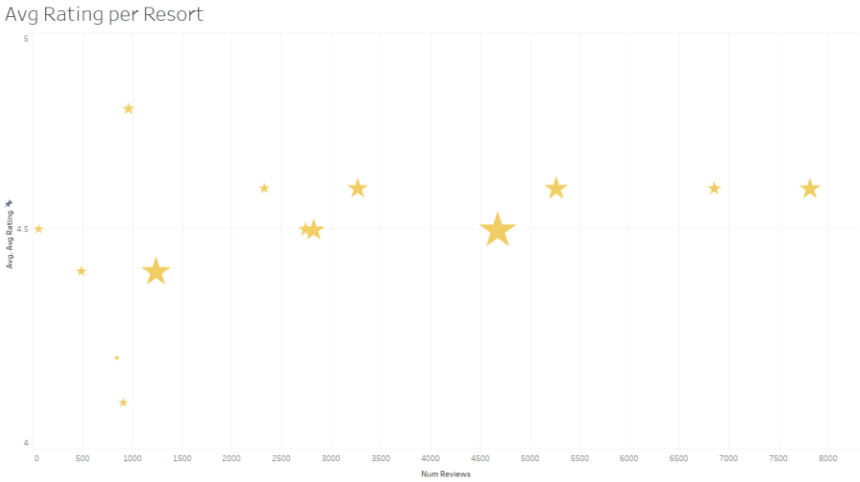
I've been having a hard time trying to use the national ski data and the Utah ski data, so I decided to narrow my scope to just the ski resorts in Utah. My audience will be skiers located or interested in visiting Utah who want to find a ski resort that best fits their preferences.

03/10/25 I really want to focus on the size and terrain distribution for each ski resort as I think these aspects will be very appealing to my audience and purpose. So, I created two different stacked bar charts that I really like, and I plan to use one on my dashboard. However, I'm having some trouble deciding whether I want to use one just using the part-of-a-whole (second chart), or implementing the total runs in the mix as well (first chart):





04/14/25 I collected data on each of the 14 Utah ski resorts from Google Reviews and added them to a new sheet on my Excel spreadsheet. I looked at the average rating per resort as well as the number of reviews. With that information, I made the visualization below. However, I'm not sure how much value this will add to my dashboard. I don't think it quite captures my purpose, and I don't think my audience will consider the number of reviews would be an important factor in choosing a ski resort. Also, all of the ski resorts have very similar ratings at around 4.5 stars.



04/15/25 After getting feedback from my classmates, I added a text block to the bottom of my dashboard containing my sources to increase credibility. I'm also considering adding captions to my visualizations and/or an explanation on the interactive features.

04/22/25 I found out how to color two columns of a highlight chart differently. This is very valuable as I can now include the price column in my dashboard without messing up the coloring! I also added the ski level column into the text part of the highlight table, and I found symbols that correspond to the different terrain/skill levels. These small changes to the highlight table add a lot more information and visual appeal to my dashboard.

Highlight Table (2)

Region	Resort Name	SkiLevel	Avg Snow	Price
Heber Valley	Sundance Mountain	◆	300.0	$119.00
Northern Utah	Beaver Mountain	● ■	400.0	$70.00
	Cherry Peak	● ■	322.0	$55.00
Ogden Valley	Nordic Valley	◆	300.0	$21.00
	Powder Mountain	■	360.0	$109.00
	Snowbasin	■	300.0	$169.00
Park City	Deer Valley	● ■	300.0	$229.00
	Park City Mountain	■ ◆	360.0	$289.00
Salt Lake	Alta	◆	546.0	$199.00
	Brighton	■	500.0	$105.00
	Snowbird	◆	500.0	$184.00
	Solitude Mountain	■ ◆	500.0	$135.00
Southern Utah	Brian Head	●	360.0	$94.00
	Eagle Point	■	350.0	$64.00